Faculty of Computers and Artificial Intelligence
Information Systems Department
Cairo University

Third Year
Data Warehouse Assignment
2026

In this assignment you should solve the below **four** questions using **Microsoft SSIS Transformations**:

1. You are provided with three different data sources: **Air_Quality.xml**, **Air_Quality.csv**, and a REST API: https://69be08c917c3d7d977910f4e.mockapi.io/us_air_quality/air_monitoring.

   Implement a data integration pipeline that consolidates air quality data from these multiple sources into a unified and clean dataset.

   Create a table named **air_quality_Q1** in the database to store the integrated data.

   During the integration process, you are required to:

   1.1 Standardize different schemas into a unified structure (sensor_id, city, timestamp, pm25, pm10).
   1.2 Handle data quality issues such as missing values, empty fields, and negative pollution readings.
   1.3 Add a column to identify the source of each record.

   Ensure the final dataset is clean, consistent, and ready for analysis.

2. Implement **SCD type 6** for the below source table ***'Campaign_Q2'***:

ID	Name	Budget	Update_Date
1001	Summer Ads	10000	1-1-2026
1002	Winter Ads	15000	1-1-2026
1003	Spring Ads	20000	1-1-2026

Notes:

1. Create the necessary **target table** with the necessary **columns**.
2. The SCD fields are **Name and Budget**.
3. Read the source data using **Incremental Load**.
4. Do not use the SCD Transformation.

3. Load source data to a target table using **versioning** like below:

3.1 Source table *'Device_Status'*

Device_ID	Device_Type	Location	Status	Schedule_Date
D101	Ventilator	ICU	Active	27/3/2026
D102	MRI	Radiology	Active	27/3/2026
D103	Pump	ER	Maintenance	27/3/2026

3.2 Target table after the <u>first</u> run on the <u>same</u> day

Device_Key	Device_ID	Device_Type	Location	Status	Insert_Date	Active_Flag	Version_No
1	D101	Ventilator	ICU	Active	27-03-2026	1	1
2	D102	MRI	Radiology	Active	27-03-2026	1	1
3	D103	Pump	ER	Maintenance	27-03-2026	1	1

3.3 Target table after the <u>second</u> run on the <u>same</u> day

Device_Key	Device_ID	Device_Type	Location	Status	Insert_Date	Active_Flag	Version_No
1	D101	Ventilator	ICU	Active	27-03-2026	0	1
2	D102	MRI	Radiology	Active	27-03-2026	0	1
3	D103	Pump	ER	Maintenance	27-03-2026	0	1
4	D101	Ventilator	ICU	Active	27-03-2026	1	2
5	D102	MRI	Radiology	Active	27-03-2026	1	2
6	D103	Pump	ER	Maintenance	27-03-2026	1	2

3.4 Target table after the first run on the next day (just change *Schedule_Date* in the source data to simulate the next day)

Device_Key	Device_ID	Device_Type	Location	Status	Insert_Date	Active_Flag	Version_No
1	D101	Ventilator	ICU	Active	27-03-2026	0	1
2	D102	MRI	Radiology	Active	27-03-2026	0	1
3	D103	Pump	ER	Maintenance	27-03-2026	0	1
4	D101	Ventilator	ICU	Active	27-03-2026	0	2
5	D102	MRI	Radiology	Active	27-03-2026	0	2
6	D103	Pump	ER	Maintenance	27-03-2026	0	2
7	D101	Ventilator	ICU	Active	28-03-2026	1	1
8	D102	MRI	Radiology	Active	28-03-2026	1	1
9	D103	Pump	ER	Maintenance	28-03-2026	1	1

Notes:

1. I have shown two runs only on the same day but during discussion, I can run **n** times and see the behavior of your solution.
2. Add **a new version** as long as you are running on the **same** day and close **all** old records.
3. Start from **version 1** again, on the **next** day and so on.
4. **Don't** check for any change in the source data, load it as it is.

4. You are a Data Engineer working for a trading analytics platform. Trade transactions are captured in a staging table and must be processed into a clean analytical format for reporting. Your task is to read data from the Raw_Trades table and produce aggregated trading metrics per trader.

4.1 Sample from the source table *'Raw_Trades'*

Trader_ID	Trade_TS	Action	Quantity
501	27/3/2026 9:00:00	BUY	100
501	27/3/2026 9:00:00	buy	100
501	27/3/2026 9:05:00	sell	50
501	27/3/2026 9:10:00	buy	-30
502	27/3/2026 9:00:00	sell	70
502	27/3/2026 9:02:00	buy	50
502	27/3/2026 9:03:00	sell	20
503	27/3/2026 9:10:00	buy	200
503	27/3/2026 9:20:00	sell	100
503	27/3/2026 9:25:00	sell	150
504	27/3/2026 9:00:00	buy	80
504	27/3/2026 9:01:00	BUY	20
504	27/3/2026 9:02:00	sell	50
505	27/3/2026 9:00:00	sell	60
505	27/3/2026 9:05:00	buy	40
506	27/3/2026 9:10:00	buy	70
506	27/3/2026 9:15:00	sell	70
506	27/3/2026 9:15:00	buy	-70
507	27/3/2026 9:20:00	buy	90
507	27/3/2026 9:25:00	sell	100

508	27/3/2026 9:30:00	buy	200
508	27/3/2026 9:35:00	buy	50
508	27/3/2026 9:40:00	sell	150
508	27/3/2026 9:45:00	SELL	-50
508	27/3/2026 9:50:00	buy	-30

4.2 Target table *'Trader_Summary'*

Att_Key	Trader_ID	Trade_Date	Total_Buy	Total_Sell
1	501	27/3/2026	100	80
2	502	27/3/2026	50	20
3	503	27/3/2026	200	250
4	504	27/3/2026	100	50
5	505	27/3/2026	40	0
6	506	27/3/2026	70	140
7	507	27/3/2026	90	0
8	508	27/3/2026	280	200
.	.	.	.	.
.	.	.	.	.

Assumptions:

1. Trade actions must be two valid values only: buy and sell
2. If a trade record contains a negative quantity, the transaction must be interpreted as the opposite action, and the quantity must be treated as a positive value.
3. A SELL transaction is considered valid only if previous Buy exists (Buy_TS < Sell_TS and Buy_Qty >= Sell_Qty)
4. Any SELL transaction that does not satisfy the above condition must be excluded from the final result.

General Notes:

1. No late submission will be accepted **for any reason**.
2. The team should consist of **TWO** students only.
3. **All** team members must attend the discussion. **ZERO** points will be given to the absentees.
4. During the discussion, I will test using **different** datasets, so be prepared for that.
5. If you have any questions, feel free to ask your TA.
6. If you complete the entire assignment **without cheating**, wallahi be sure that you have **learned** a lot and you have done something **great**.

Where to submit?

1. Prepare a **zip** file that contains four folders, one for each problem, and name it as follows:
   DWH_Assignment_TA_Name_ID1_ID2.zip

   Ex: DWH_Assignment_Nehal_20220011_20220022.zip

2. Upload the zip file to this Google Form: https://forms.gle/LpJQ51NoQrbgn9et8

3. The form will be closed on **Thursday, April 16th at 11:59 PM**.

Wishing you all the best ❤